Latent Landscapes

When can ecological process be recovered from imperfect spatial observations?

Synthetic methods -> mink-water vole -> Australian predator impacts

Barry W. Brook

We reset because real-data failure was impossible to diagnose

01

Discovery concept

Process, observation and inference sketched at a high level

02

Mink-water vole start

Realism arrived before parameter truth or diagnostic control

03

Known-truth synth-small

Synthetic control helped, but the wave was too simple

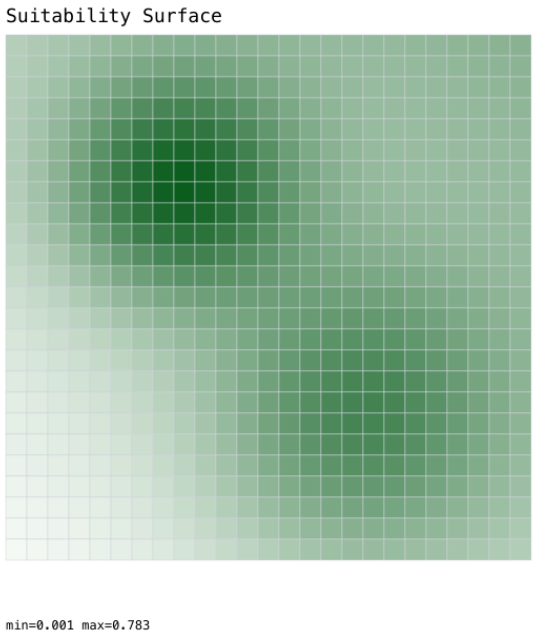
04

Latent Landscapes

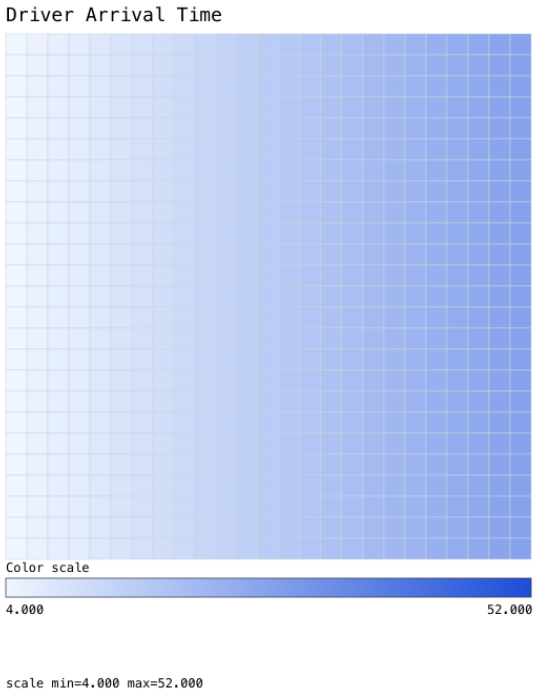
A richer Python laboratory isolates one failure mode at a time

Progressive diagnosis, not a straight-line success story.

The first synthetic model was understandable - but not informative enough



Smooth habitat



Near-linear arrival wave

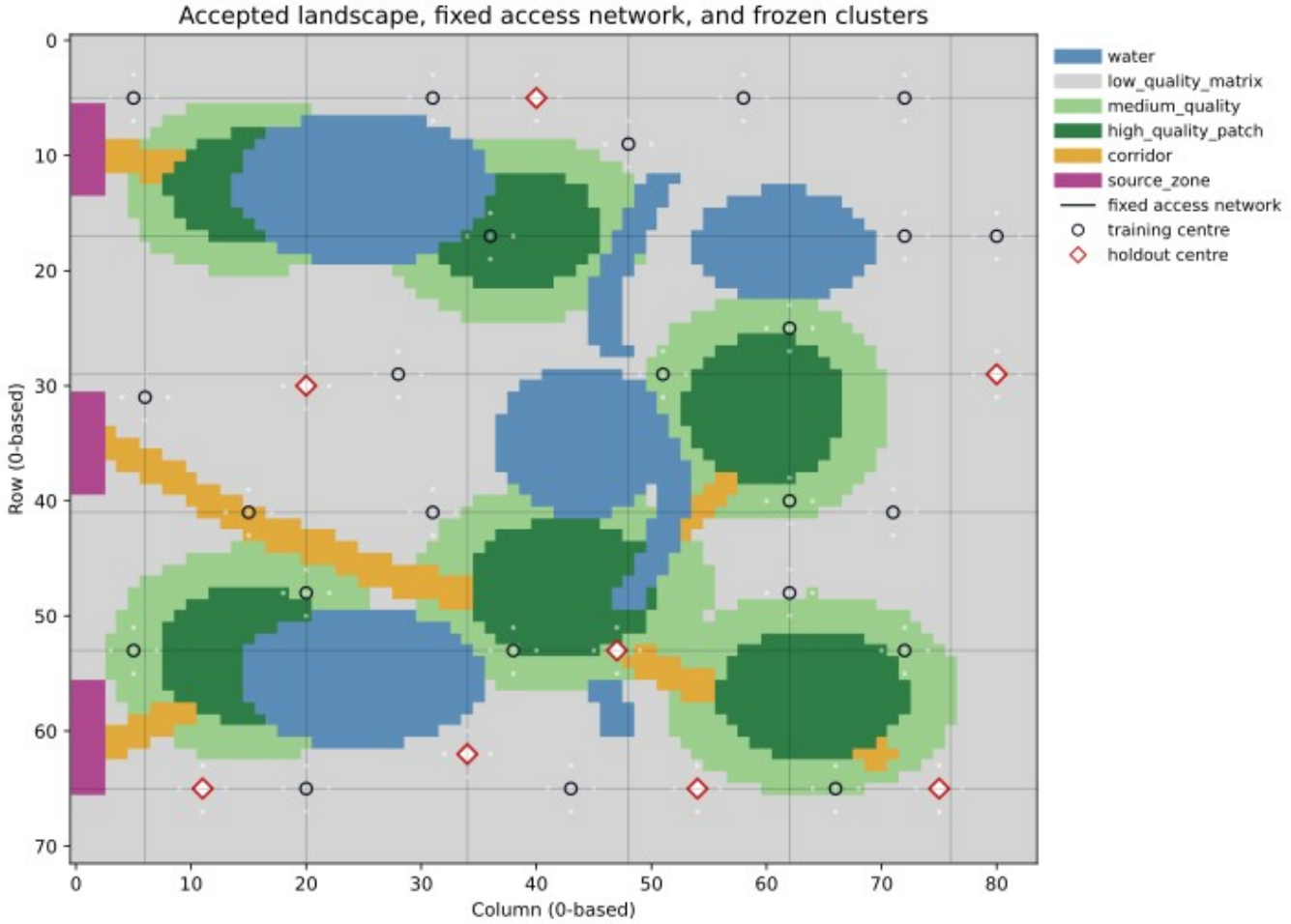
What it gave us

- Known truth and easy visual checks
- A safe place to expose observation failures

What it could not give us

- Enough spatial contrast to distinguish mechanisms
- A realistic test of resistance, barriers or biased sampling

The current laboratory makes resistance and biased sampling explicit



72 x 84 cells

- Water barriers; passable low-quality matrix
- High-conductance corridors and high-quality patches
- Three western predator source zones

Fixed road / transect network

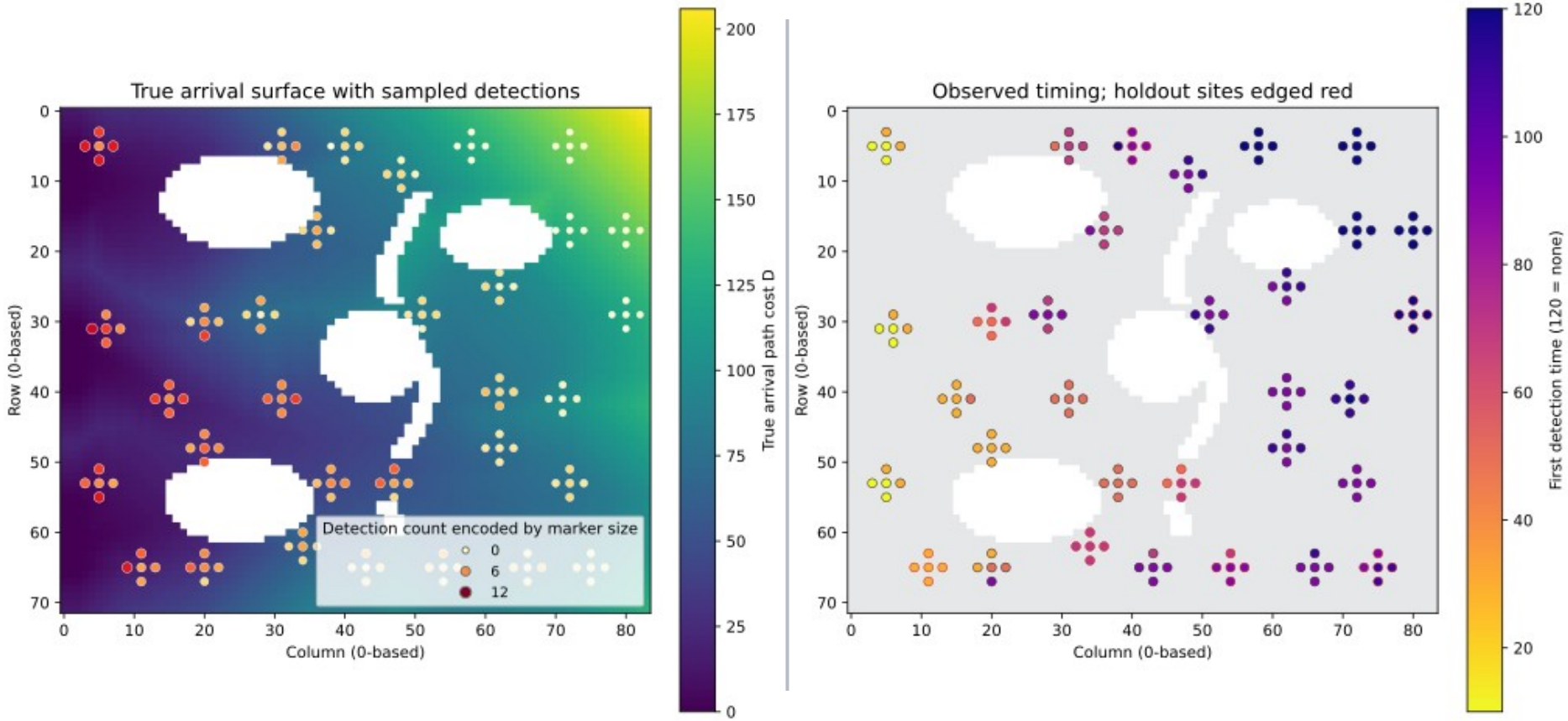
32 clusters; 8 held out

Clusters are balanced by distance from sources - not targeted to good habitat.

Imperfect detection fragments the latent spread history

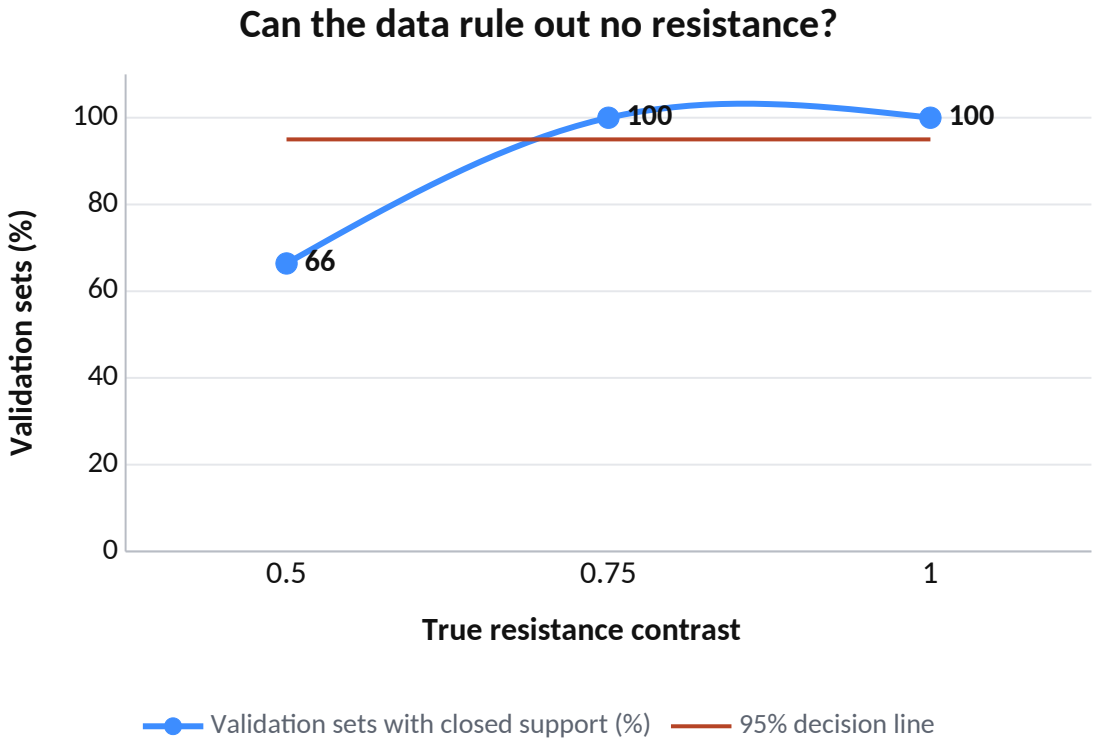
LEFT colour = true arrival; marker size + hue = total detections

RIGHT colour = first detection; purple = never detected; red edge = holdout

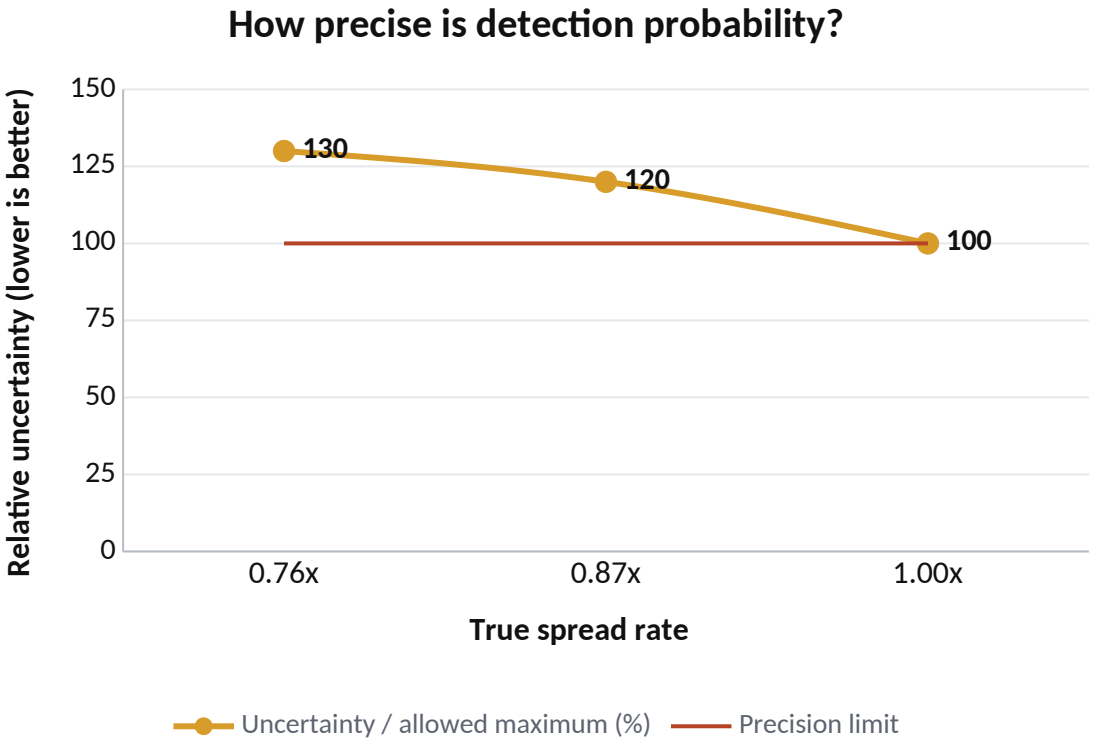


A non-detection is not an absence: detection probability is 0.65 over repeated surveys.

Ecological signal has thresholds: weak patterns can be real yet unrecoverable

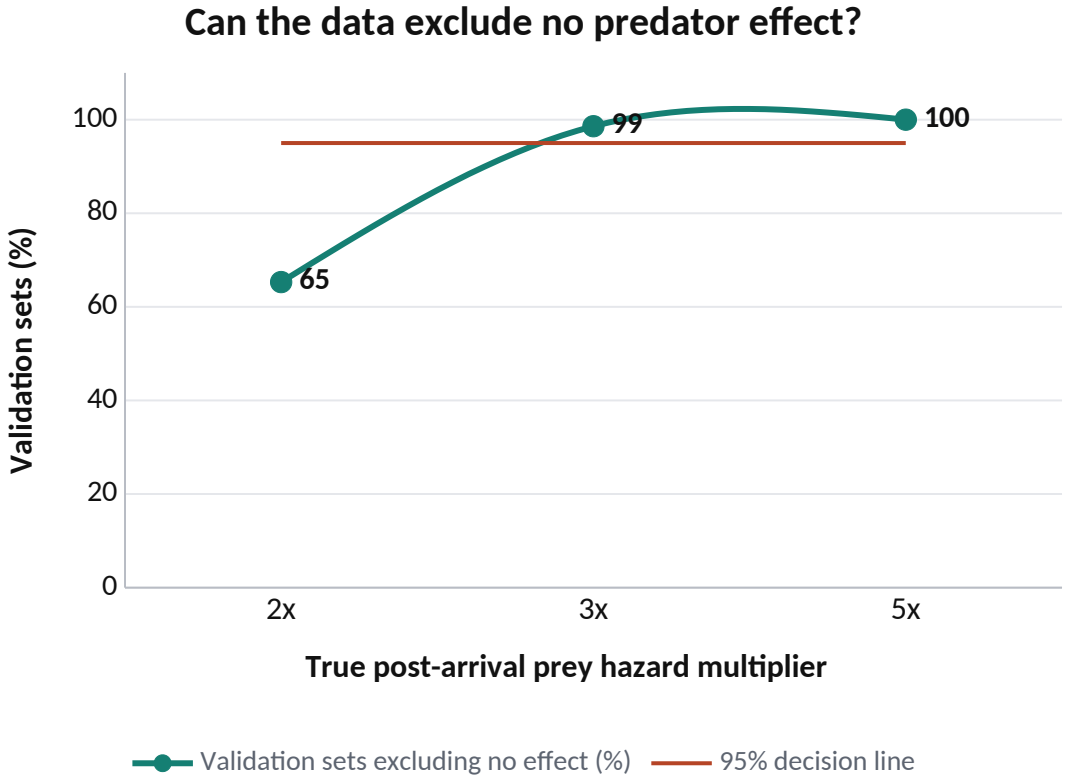


At contrast 0.5, one-third of datasets still admit uniform resistance.



Slower waves estimate spread well, but leave detection probability too broad.

The prey signal also has a strength threshold - then survives observation noise



1 Known predator arrival

5x prey effect recovered

2 Imperfect prey detection

Recovery retained at p = 0.65

3 Latent predator arrival

Recovery retained after integrating arrival uncertainty

For this design, effect strength mattered more than perfect observability.

Joint calibration obscured prey information the data retained

ONE SIMULTANEOUS REGION

13% of validation datasets excluded no prey effect

1x - 10x typical prey-effect region filled the tested grid

Point estimates for spread and prey effect were nevertheless centred on truth.

ONE INTERVAL PER PARAMETER

100% of validation datasets excluded no prey effect

~2-fold typical prey-effect interval width

Separate spread and prey intervals each covered truth in about 95% of datasets.

Same model and data; a different uncertainty question. Simultaneous joint coverage is still unresolved.

Identifiability belongs to the whole experiment - not to a parameter alone

OBSERVED

Ecological signal

Weak resistance and weak prey effects can be real yet indistinguishable from simpler explanations.

OBSERVED

Observation design

Slow waves and imperfect detections leave less information for separating process from detection.

INFERENCE

Uncertainty calibration

A joint cutoff can hide parameter-specific information even when the estimates themselves are sound.

Still untested **simultaneous joint coverage | new landscapes | model misspecification | synth-realistic stress test**

Rigour at the scientific decision boundary; proportional engineering everywhere else.

Three papers add ecological realism by design

1

Synthetic methods

Known truth; identifiability thresholds; calibrated uncertainty; more than one landscape/model.

2

Mink-water vole

Great Britain as a constrained real-world test; islands retained as potential natural contrasts.

3

Australian murids

Cats and potentially foxes under sparse, difficult data; the grant-scale ecological endpoint.

Each paper earns the next increase in realism.

Three questions for the team

1. Which synthetic contrasts would make the identifiability paper most ecologically persuasive?

2. What is the minimum extra landscape / model set needed to avoid a one-case story?

3. Which constraints from mink-water vole and Australian data should shape the synthetic work now?